PHYSICS FORMULA & METHODS HANDBOOK

CHAPTER 6: SYSTEMS OF PARTICLES AND ROTATIONAL MOTION | CBSE / JEE / NEET
EXAM-ORIENTED NOTES

1. Centre of Mass (CM)

TOPIC: POSITION VECTOR OF CENTRE OF MASS

$$\vec{R}_{\text{cm}} = \frac{\sum_{i=1}^{n} m_i \vec{r}_i}{M} = \frac{\int \vec{r}\, dm}{M}$$

Discrete Components: $X_{\text{cm}} = \frac{\sum m_i x_i}{M}, \quad Y_{\text{cm}} = \frac{\sum m_i y_i}{M}, \quad Z_{\text{cm}} = \frac{\sum m_i z_i}{M}$
Continuous Components: $X_{\text{cm}} = \frac{1}{M}\int x\, dm, \quad Y_{\text{cm}} = \frac{1}{M}\int y\, dm, \quad Z_{\text{cm}} = \frac{1}{M}\int z\, dm$

TOPIC: KINEMATICS & DYNAMICS OF CENTRE OF MASS

$$\vec{V}_{\text{cm}} = \frac{d\vec{R}_{\text{cm}}}{dt} = \frac{\sum m_i \vec{v}_i}{M} = \frac{\vec{P}_{\text{total}}}{M}$$

$$\vec{A}_{\text{cm}} = \frac{d\vec{V}_{\text{cm}}}{dt} = \frac{\sum m_i \vec{a}_i}{M} = \frac{\vec{F}_{\text{ext}}}{M}$$

Conservation of Momentum: If $\vec{F}_{\text{ext}} = 0 \implies \vec{P}_{\text{total}} = \text{constant}$ and $\vec{V}_{\text{cm}} = \text{constant}$.

SHORTCUT: MASS REMOVAL / CUT-OUT BODIES

When a shape of mass m' is removed from a uniform object of mass M_0, the new centre of mass relative to original origin is:

$$X_{\text{new}} = \frac{M_0 X_0 - m' x'}{M_0 - m'}$$

2. Vector Product & Kinematics of Rotation

TOPIC: VECTOR (CROSS) PRODUCT

$$\vec{A} \times \vec{B} = (AB\sin\theta)\hat{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

Properties: Anti-commutative ($\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$). Unit vectors: $\hat{i} \times \hat{j} = \hat{k}$, $\hat{j} \times \hat{k} = \hat{i}$, $\hat{k} \times \hat{i} = \hat{j}$.

TOPIC: RELATION BETWEEN LINEAR AND ANGULAR VARIABLES

$$\vec{v} = \vec{\omega} \times \vec{r} \implies v = \omega r_{\perp}$$

$$\vec{a}_t = \vec{\alpha} \times \vec{r} \implies a_t = \alpha r_{\perp} \quad \text{and} \quad a_c = \frac{v^2}{r} = \omega^2 r$$

Angular Kinematic Equations (Constant α):

$$\omega = \omega_0 + \alpha t \quad | \quad \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \quad | \quad \omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0)$$

3. Torque & Angular Momentum

TOPIC: TORQUE (MOMENT OF FORCE)

$$\vec{\tau} = \vec{r} \times \vec{F} \implies \tau = rF\sin\theta = r_{\perp}F = rF_{\perp}$$

SI Unit: $\mathrm{N\ m}$ | **Dimensions:** $[ML^2T^{-2}]$
Rotational Form: $\vec{\tau}_{\text{ext}} = I\vec{\alpha}$ (for rotation about a fixed axis).

TOPIC: ANGULAR MOMENTUM (MOMENT OF LINEAR MOMENTUM)

$$\vec{L} = \vec{r} \times \vec{p} \implies L = rp\sin\theta = r_{\perp}p$$

$$\vec{L}_z = I\vec{\omega} \quad (\text{for symmetric rigid bodies rotating about fixed z-axis})$$

SI Unit: $\mathrm{J\ s}$ or $\mathrm{kg\ m^2 s^{-1}}$ | **Dimensions:** $[ML^2T^{-1}]$
Fundamental Dynamic Law: $\vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}$

MUST KNOW: LAW OF CONSERVATION OF ANGULAR MOMENTUM

If the net external torque acting on a system is zero ($\vec{\tau}_{\text{ext}} = 0$):

$$\frac{d\vec{L}}{dt} = 0 \implies \vec{L} = \text{constant} \implies I_1\omega_1 = I_2\omega_2$$

Application: A spinning acrobat folding arms reduces I, causing angular speed ω to increase.

4. Moment of Inertia (I) & Energy

TOPIC: MOMENT OF INERTIA & RADIUS OF GYRATION

$$I = \sum_{i=1}^{n} m_i r_i^2 = \int r^2\, dm \quad \text{and} \quad I = Mk^2 \implies k = \sqrt{\frac{I}{M}}$$

SI Unit: $\mathrm{kg\,m^2}$ | **Dimensions:** $[ML^2]$
Rotational Kinetic Energy: $K_{\text{rot}} = \frac{1}{2}I\omega^2 = \frac{L^2}{2I}$
Work & Power: $W = \int \tau\, d\theta, \quad P = \vec{\tau} \cdot \vec{\omega}$

5. Standard Moments of Inertia Reference Table

Body	Axis of Rotation	Moment of Inertia (I)	Radius of Gyration (k)
Thin Circular Ring (Radius R)	Perpendicular to plane at centre	MR^2	R
Thin Circular Ring (Radius R)	Along any diameter	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
Thin Rod (Length L)	Perpendicular to rod at mid-point	$\frac{1}{12}ML^2$	$\frac{L}{\sqrt{12}}$
Circular Disc (Radius R)	Perpendicular to disc at centre	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
Circular Disc (Radius R)	Along any diameter	$\frac{1}{4}MR^2$	$\frac{R}{2}$
Hollow Cylinder (Radius R)	Central axis of cylinder	MR^2	R
Solid Cylinder (Radius R)	Central axis of cylinder	$\frac{1}{2}MR^2$	$\frac{R}{\sqrt{2}}$
Solid Sphere (Radius R)	Along any diameter	$\frac{2}{5}MR^2$	$\sqrt{\frac{2}{5}}R$

6. Equilibrium & Levers

TOPIC: CONDITIONS FOR MECHANICAL EQUILIBRIUM

$$\text{Translational Equilibrium: } \vec{F}_{\text{net}} = \sum \vec{F}_i = 0 \implies \vec{P} = \text{constant}$$

$$\text{Rotational Equilibrium: } \vec{\tau}_{\text{net}} = \sum \vec{\tau}_i = 0 \implies \vec{L} = \text{constant}$$

TOPIC: PRINCIPLE OF MOMENTS (LEVER RULE)

$$F_1 d_1 = F_2 d_2 \implies \text{Load} \times \text{Load Arm} = \text{Effort} \times \text{Effort Arm}$$

$$\text{Mechanical Advantage (M.A.)} = \frac{F_1}{F_2} = \frac{d_2}{d_1}$$

7. Translational vs Rotational Motion Analogies

Translational Kinematics / Dynamics	Rotational Kinematics / Dynamics
Linear Displacement (x)	Angular Displacement (θ)
Linear Velocity ($v = \frac{dx}{dt}$)	Angular Velocity ($\omega = \frac{d\theta}{dt}$)
Linear Acceleration ($a = \frac{dv}{dt}$)	Angular Acceleration ($\alpha = \frac{d\omega}{dt}$)
Mass (M)	Moment of Inertia (I)
Force ($F = Ma$)	Torque ($\tau = I\alpha$)
Work ($dW = F\,dx$)	Work ($dW = \tau\,d\theta$)
Kinetic Energy ($K = \frac{1}{2}Mv^2$)	Kinetic Energy ($K = \frac{1}{2}I\omega^2$)
Power ($P = Fv$)	Power ($P = \tau\omega$)
Linear Momentum ($P = Mv$)	Angular Momentum ($L = I\omega$)

8. Numerical Problem-Solving Methods

Derivation Method: Deriving Rotational Kinematic Equation $\omega^2 = \omega_0^2 + 2\alpha\theta$

By definition of angular acceleration: $\alpha = \frac{d\omega}{dt} = \frac{d\omega}{d\theta} \cdot \frac{d\theta}{dt} = \omega\frac{d\omega}{d\theta}$

Rearranging variables for integration:

$$\omega\,d\omega = \alpha\,d\theta$$

$$\int_{\omega_0}^{\omega} \omega\,d\omega = \alpha \int_0^{\theta} d\theta$$

$$\left[\frac{\omega^2}{2}\right]_{\omega_0}^{\omega} = \alpha\theta \implies \frac{\omega^2 - \omega_0^2}{2} = \alpha\theta$$

$$\boxed{\omega^2 = \omega_0^2 + 2\alpha\theta}$$

Numerical Solver: Flywheel Torque & Work Done Step-by-Step

$$\text{Given Values} \rightarrow \text{Select Equation} \rightarrow \text{Substitute} \rightarrow \text{Calculate} \rightarrow \text{Final Answer}$$

Problem Example: A solid disc flywheel of mass $M = 20\ \text{kg}$ and radius $R = 0.2\ \text{m}$ is pulled with a tangentially applied force $F = 25\ \text{N}$. Find angular acceleration α and kinetic energy gained after unwinding $s = 2\ \text{m}$ of string.

Step 1: Moment of Inertia of Disc: $I = \frac{1}{2}MR^2 = \frac{1}{2}(20)(0.2)^2 = 0.4\ \text{kg m}^2$

Step 2: Applied Torque: $\tau = F \times R = 25 \times 0.2 = 5\ \text{N m}$

Step 3: Angular Acceleration: $\alpha = \frac{\tau}{I} = \frac{5}{0.4} = \boxed{12.5\ \text{rad s}^{-2}}$

Step 4: Angular Displacement: $\theta = \frac{s}{R} = \frac{2}{0.2} = 10\ \text{rad}$

Step 5: Work-Energy Theorem Verification: $W = \tau\theta = 5 \times 10 = \boxed{50\ \text{J}}$ (Equal to $K_{\text{rot}} = \frac{1}{2}I\omega^2$).